Iqra Irfan

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Technical Skills

Languages: Python, Java, TypeScript/JavaScript, C, C#, C++
Data Analysis/Databases: PostgreSQL, MongoDB, MySQL; Query Processing/Optimization, NoSQL
Finance/Data: Bloomberg Terminal, Excel Modeling, Financial Data Analysis, Investment Reporting Automation, Structured and Unstructured Data Processing
Backend: Flask, FastAPI, Node.js, Next.js, .NET, REST APIs
ML/AI Systems: Claude/Claude Code, OpenAI APIs, TensorFlow, Reinforcement Learning, Query Optimization
Cloud/DevOps: AWS, Docker, Kubernetes, Git, BitBucket

Experience

Software Analyst Sept. 2024 – Nov. 2025
Cliffwater *Marina del Rey, CA*

- Promoted from intern to full-time Software Analyst, developing AI-assisted automation workflows, internal financial tooling, API integrations, and secure data-processing systems supporting investment operations.
- Built and maintained production automation systems that generated and distributed daily financial reporting for 300+ investors while securely processing encrypted financial data.
- Led development efforts across multiple internal tools, coordinating task execution and improving operational reliability with zero production-critical failures.
- Used Python, SQL, cloud infrastructure, and modern AI tooling to streamline workflows, reduce manual operational overhead, and support data-driven investment analysis.

Junior Full-Stack Mobile Software Engineer Jan. 2023 – Sept. 2024
7000 Languages *Los Angeles, CA*

- Promoted from spring intern to junior-level full-stack mobile application software engineer after helping build a cross-platform iOS/Android app for indigenous language learning using React Native, TypeScript, and Python.
- Introduced a dynamic activity bar and interactive home page, and led UI refactoring efforts that improved performance, usability, and overall user experience.
- Improved app stability by implementing offline data storage with MongoDB/Realm in an agile team environment, contributing to positive user feedback and increased learning outcomes.

Projects

Automated Private Investments Daily Financial Statement Production (Cliffwater) Aug. 2025

- Developed a C# automation system to generate and distribute daily investor statements for three major funds. Integrated UMB data retrieval, encrypted SQL storage, and PDF generation via HTML, automated through Docker, Kubernetes, Azure logging, and OpsGenie alerts.
- Impacted 300+ direct investors while handling highly sensitive financial and legal information.

DataIQ: Database Mapper And Intelligence Tool Jan. 2026

- Building a database intelligence platform for schema mapping, dependency analysis, and workload-aware optimization insights across evolving production databases; focused on surfacing query performance bottlenecks and safe refactoring opportunities for data engineers.
- Designing a PostgreSQL-first MVP that extracts system catalog metadata, analyzes query patterns/execution signals, models tables/joins/indexes as an interactive graph, and flags unused columns, redundant tables, missing indexes, and expensive joins; expanding support to MSSQL and Snowflake/Redshift-style warehouse schemas. Written in C++.

BaoLite: A DQN-Based Learned Join Order Optimizer for PostgreSQL Oct. 2025

- Built a reinforcement learning–based join-order optimizer for PostgreSQL using planner hooks and a Deep Q-Network (DQN) policy, inspired by Bao (VLDB '19).
- Automated extraction of 6,191+ query plans and benchmarked BaoLite against PostgreSQL's native planner on Stack workloads to evaluate join quality and planning latency. Produced performance gains up to 11x on large-table joins. This project inspired my next project, DataIQ.

Education

University of Southern California Los Angeles, CA
Masters - Computer Science *Jan. 2025 – Dec. 2026*

California State University - Los Angeles Los Angeles, CA
Post-Baccalaureate - Computer Science *Sept. 2022 – Dec. 2024*

University of California, Los Angeles Los Angeles, CA
Bachelors of Science - Human Biology and Society, ***Highest Departmental Honors*** *Jul. 2018 – May 2022*